Language, Emotion, and Collective Action

Evidence From Wisconsin Women's Organizations on Facebook

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Introduction

- Social media platforms play a central role in how organizations communicate with the public.
- Online engagement (likes, shares, comments) reflects public attention, amplification, and interaction.
- Emotional expressions and linguistic features in posts may influence engagement levels.
- This project analyzes Facebook post messages using sentiment analysis, emotion classification, topic modeling, and regression modeling.
- **Goal:** Understand how language features and discrete emotions relate to online and offline engagement outcomes.

Research Questions

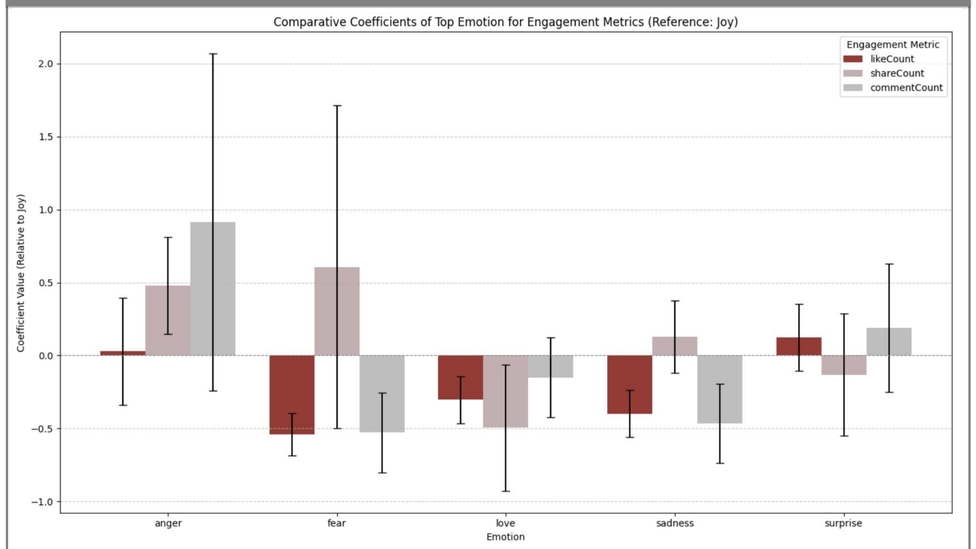
RQ1. How do different emotional expressions in Facebook posts relate to online engagement outcomes (likes, shares, comments)?

RQ2. What are the prevalent topics posted by feminist organizations, and how is this connected with online and offline engagement (protests, rallies, marches)?

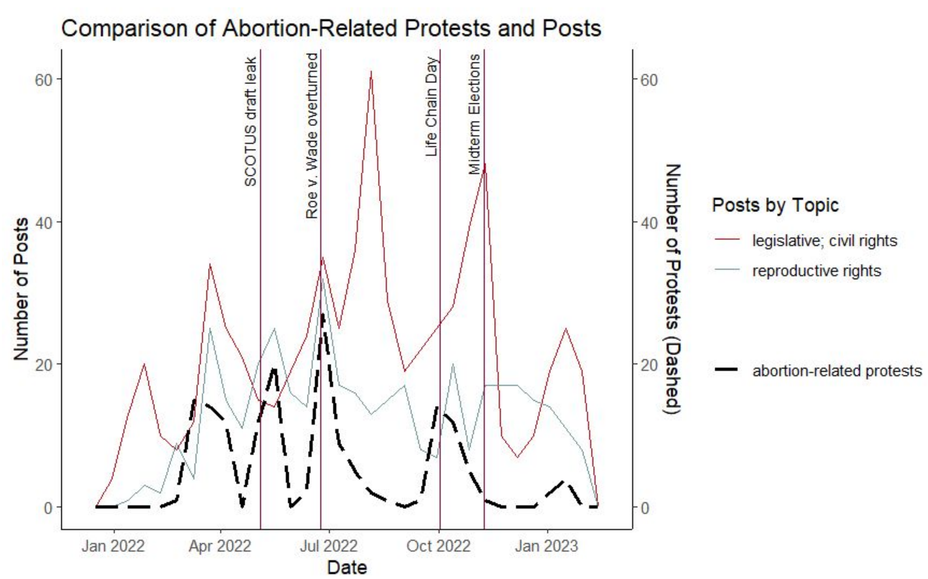
Methods

- Collected and cleaned Facebook post text and engagement data.
- **Emotion Classification** using GoEmotions model.
- **Structural Topic Modeling** to identify themes.
- **Negative Binomial Regression** to test effects of emotions on likes, shares, and comments (controlling for subscriber count and date).

Results



- **Joy** functions as a neutral, moderate-engagement baseline.
- **High-arousal emotions (anger, fear)** are the strongest drivers of engagement.
- **Low-arousal emotions (love, sadness)** consistently dampen engagement.



- Comparison of topic-specific activism during narrowed 2022 window
- **Both online and offline activity spike together** at major events
  - Posts tagged as "legislative" show mildly divergent trends

Discussion

- **Emotional expressions meaningfully shape how audiences respond to social media posts.**
  - **Joy** is generally the most effective emotion for generating likes and comments.
  - **Anger** is uniquely effective at driving shares, consistent with mobilization theories.
  - **Lower-arousal emotions** such as sadness decrease engagement across all metrics.
- **Topic modeling suggests emergent post topics are often a function of the feminist organization's niche**
  - **Top topics range from general** (abortion, health, voting) **to specific** (local ecology, youth run club, Black fitness) as a result
- Date-mapped online and offline activity for similar topics suggest that the two are linked
  - May be developed into signal for social media mobilization theories
  - Future time-series analysis may show correlational relationships

Documentation

GitHub repository with code for replicating graphs, detailed results, and references.

Acknowledgments

We thank Luhang Sun for providing the essential dataset access and insightful mentorship